Singularity Health Center

Delivering a hyperscale agent platform through rigorous architectural and operational discipline.

Presenter: Bharat
Role: Senior Engineering Manager, Agent Platform
Context: Executive Technical Review

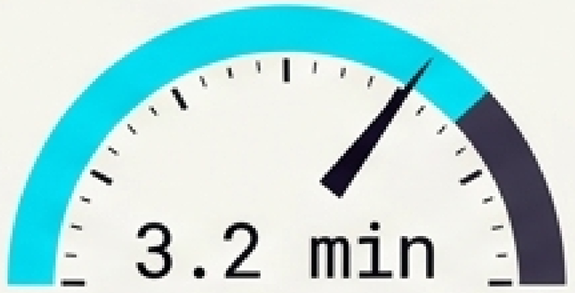
Delivering a hyperscale platform while managing live operational load

The real challenge is not just processing **billions of events**. It is landing a new **strategic platform** while maintaining an existing legacy system that is actively creating customer pain.

Operational Posture

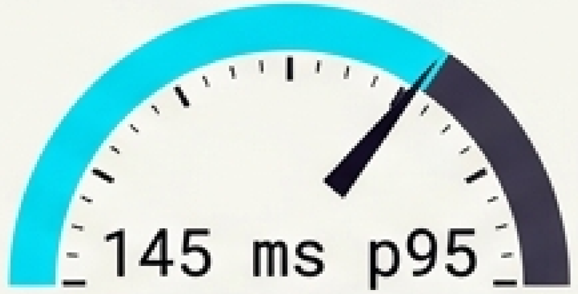
1. Protect customers now.
2. Reduce legacy drag.
3. Land the new platform through staged scope and evidence-based decisions.

Detection Freshness



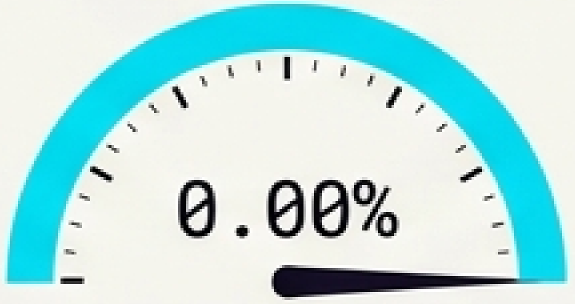
Target: < 5 min

API Read Latency



Target: < 200ms p95

Event Loss Tolerance



Target: 0 after durable bus

Isolating the real-time detection path from dashboard query patterns

Filtration Funnel

Billions of Raw Events / Millions of Agents

↓

Central Ingestion Gateway

↓

Durable Event Bus (Kafka / PubSub / Kinesis)

↓

Stream Processing (Flink / Kafka Streams)
+ Rules + Coalescing Engine

↓

Operational Alert Store (DynamoDB / Cassandra)
& Read Model + Search Index (OpenSearch)

↓

APIs + Console

↓

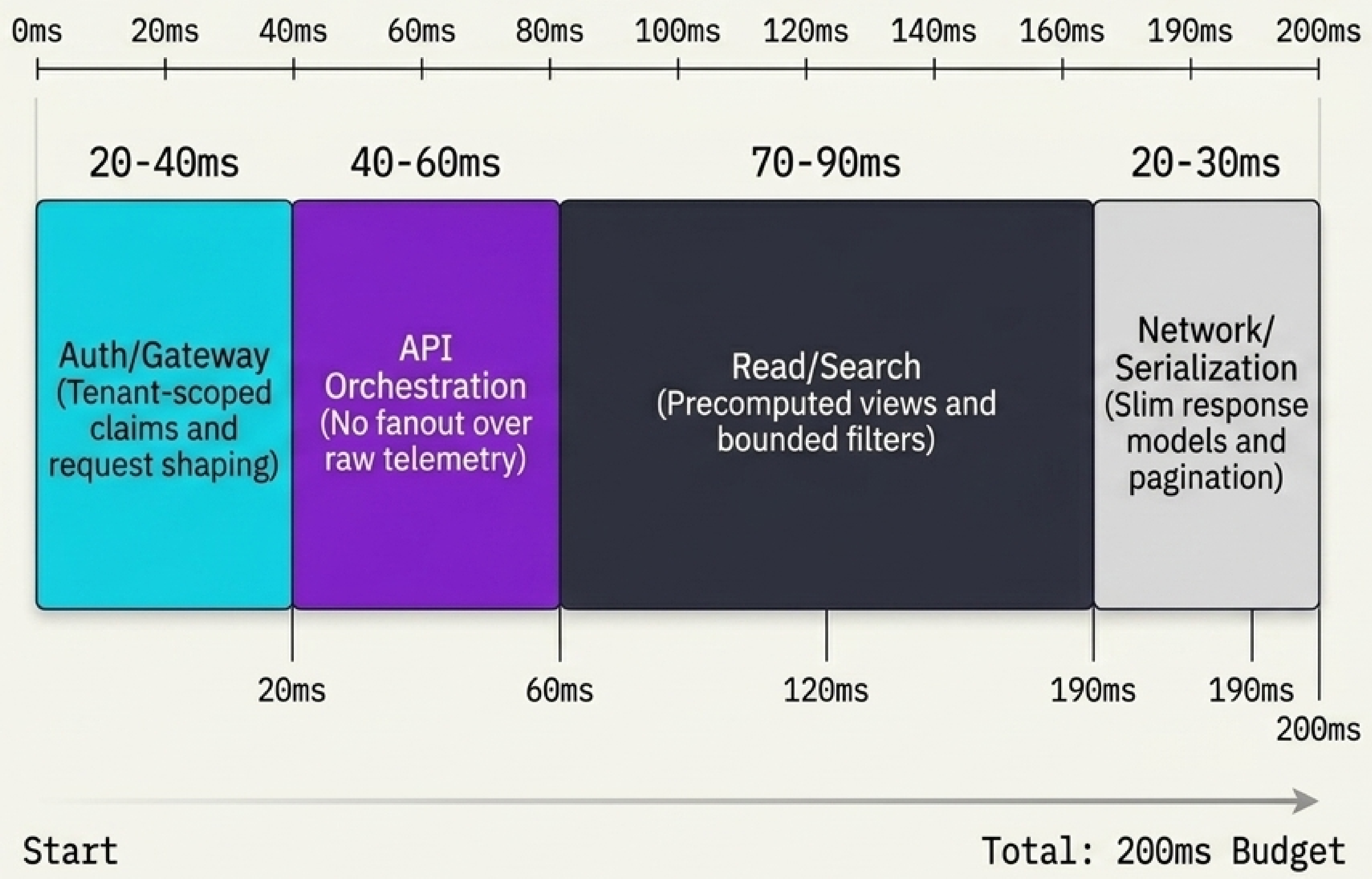
Data Lake

Hot Path: Event-time detection, coalescing, and freshness SLO enforcement.

Read Path: Precomputed dashboard views and bounded queries.

Cold Path: Long-term retention, audit, replay, and rule tuning.

Defending the <200ms p95 dashboard API latency requirement



Strict Query Discipline

- Use cursor pagination for alert lists.
- Cap time windows; no tenant-wide scans from UI paths.
- Graceful Degradation: If search lags, serve current alert state with explicit freshness metadata.

Resolving the coalescing architecture conflict through evidence

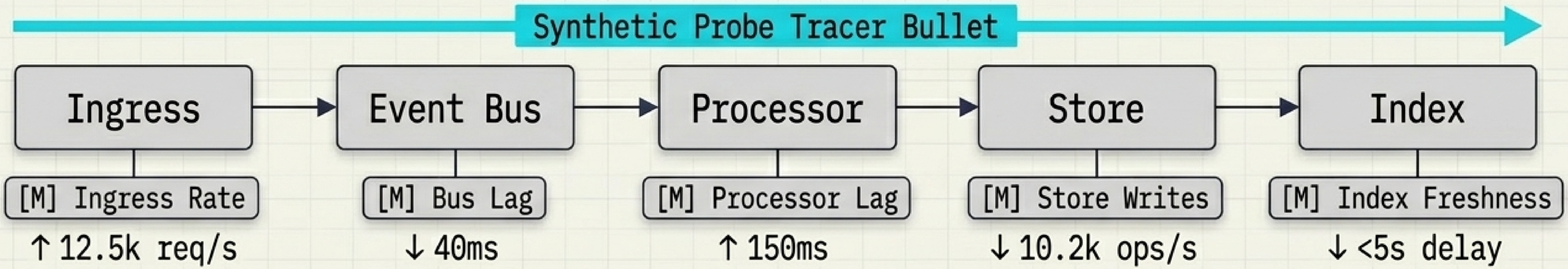
Diagnostic Comparison Table

	Engineer A (Stream Processing / Flink)	Engineer B (DB writes + Cron)
Freshness (<5m SLO)	●	○
Scalability	●	○
Infrastructure Complexity	◐	●
Replay Semantics	●	○

The Resolution Framework

1. **Criteria:** Define freshness, scale, cost, and operability constraints.
2. **Spike:** Run a one-week design spike with realistic cardinality and failure cases.
3. **Decision:** ADR commits to Stream Processing for MVP due to the strict 5-minute detection requirement.
4. **Commitment:** Eng A leads processing design; Eng B leads simplicity controls (bounded scope, runbooks, rollback plan). Both own the final result.

Detecting and mitigating the silent processing stop



> A **stale green** Health Center is worse than an **explicit degraded state**.

Sev-1 Execution Timeline

Declare: Single IC, establish customer impact clock

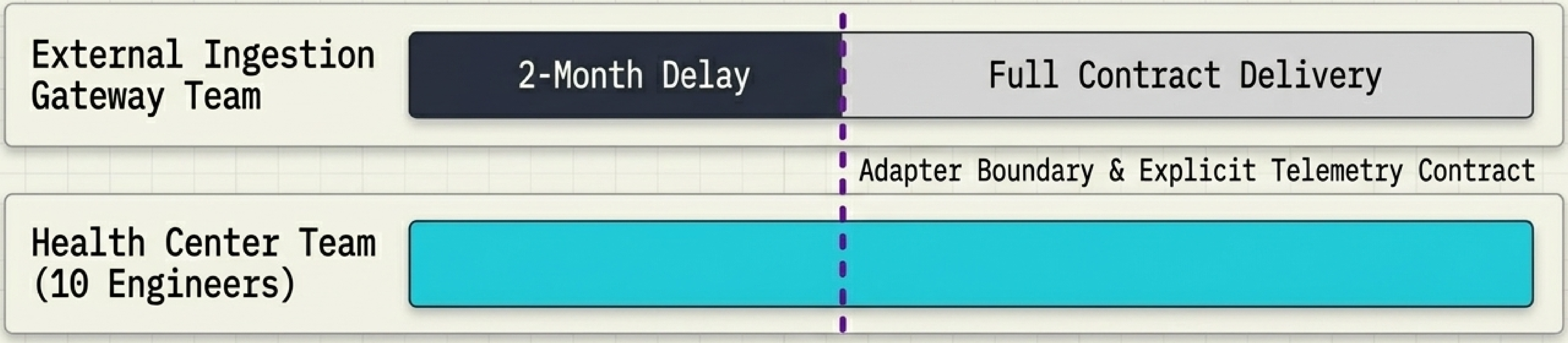
→ **Isolate:** Compare stage metrics to determine blast radius by tenant/region

→ **Mitigate:** Failover consumers, pause bad rules, or disable rollout.

→ **Recover:** Replay from durable offsets and track backfill completion.

NotebookLM

Preserving forward momentum despite a two-month gateway delay



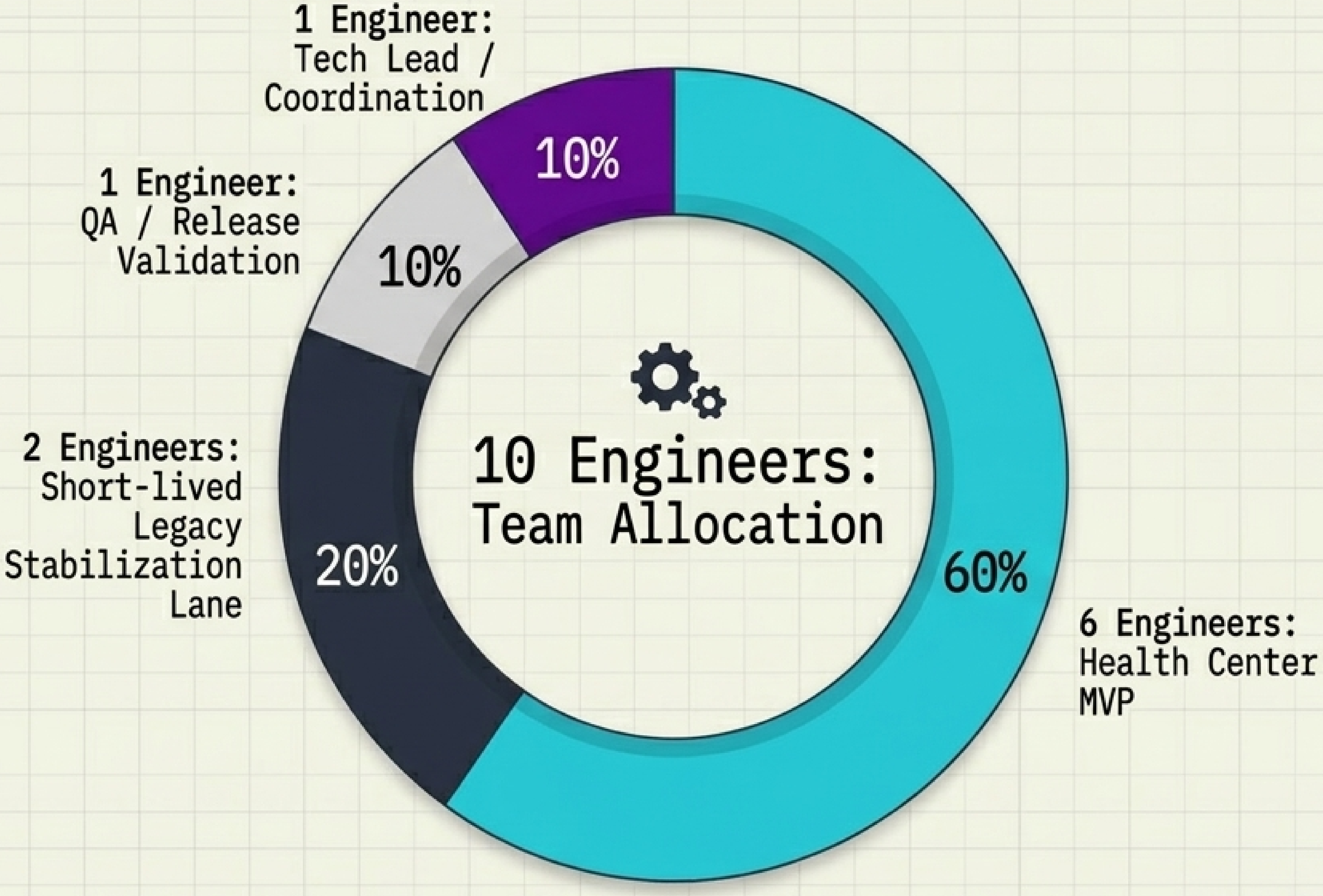
Tactical Execution Panel

Unblock Downstream: Generate synthetic and replayed telemetry streams to unblock processors, APIs, UI, and QA.

Thin-Slice Integration: Negotiate routing for minimal, high-value signals before the full contract arrives.

Leadership Rule: Teams should never become idle because an external dependency slips.

Containing the 40% legacy escalation spike without derailing the MVP



The Strategy

Treat the 40% spike in false “Offline” status as a customer trust risk, not background maintenance. Instrument the failure and ship a narrow patch.

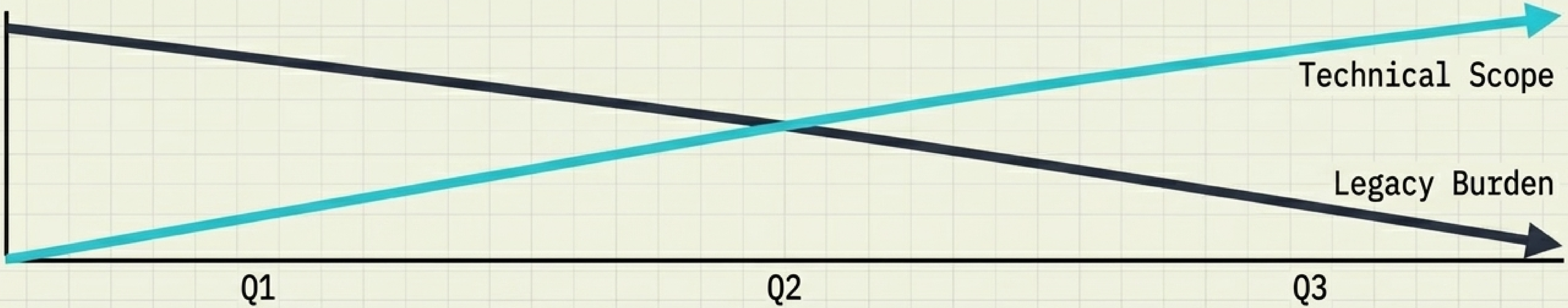
Exit Criteria

Legacy lane ends after top root cause patch and support-volume trend review. Root causes become Health Center test cases.

Escalation Rule

Increase legacy allocation only if issues cause Sev-1/Sev-2 impact across major tenants.

Evolving from MVP to a reusable Agent Platform capability



Q1: Foundations and MVP

- Top 4 anomaly rules (Offline, Disabled, Anti-tamper, Low Disk).
- Ingestion adapter, stream skeleton, alert store.
- Tenant allowlist rollout.

Gate: End-to-end synthetic telemetry operational.

Q2: GA Readiness

- Harden scale, replay, and idempotency.
- Alert lifecycle, suppression, richer search.
- Shadow-compare Health Center against legacy heartbeat.

Gate: False-positive/negative rates mapped per tenant.

Q3: Platformization

- Migrate selected offline logic to Health Center.
- Formalize reusable coalescing framework and platform APIs.
- Retire safe legacy paths.

Gate: Support burden reduced, decommission runbooks executed.

Securing customer trust and a durable platform foundation

The Inherent Risks

- Stream processing complexity.
- Gateway dependency slips.
- Legacy escalations consuming roadmap capacity.

Engineered Mitigations

- Constrained MVP rules, explicit windows, and rollback flags.
- Adapter contracts and synthetic data streams.
- Dedicated stabilization lane with strict exit criteria.

Expected Business Outcomes

- Faster, highly trustworthy agent-health visibility (< 5 min).
- A 50% reduction in support burden from false offline states.
- A reusable, durable operational intelligence foundation for all future Agent Platform use cases.